

PROJECT DESIGN PHASE-I

Problem - Solution Fit Template

DATE	12 July 2026
TEAM ID	
PROJECT NAME	Plugging into the Future: An Exploration of Electricity Consumption Patterns Using Tableau
MAXIMUM MARKS	2 Marks

Problem - Solution Fit Analysis:

The Problem-Solution Fit phase ensures that the identified customer problems are effectively addressed by the proposed solution. In this project, the major challenge was analyzing large electricity consumption datasets using manual methods. To overcome this, an interactive Tableau dashboard was designed to provide clear visual insights and dynamic filtering options. The solution directly aligns with user needs by simplifying data interpretation and improving decision-making efficiency.

PROBLEM AREA	IDENTIFIED PROBLEM	PROPOSED SOLUTION	HOW THE SOLUTION FITS
Data Interpretation	Electricity consumption data is stored in raw Excel sheets and difficult to interpret.	Developed interactive Tableau dashboard with visual charts and graphs.	Converts raw data into clear visual insights for easy understanding.
Year-wise Comparison	Difficult to compare electricity usage between 2019 and 2020.	Created Year-wise bar charts and line charts for comparison.	Enables quick and clear comparison of yearly consumption trends.
Regional Analysis	No clear visibility of region-wise electricity distribution.	Designed pie chart for total region consumption.	Helps identify which region consumes the most electricity.
State Ranking	Hard to identify top and bottom consuming states.	Implemented Top N and Bottom N analysis using parameters.	Allows dynamic identification of high and low performing states.
Trend Identification	Monthly consumption patterns are unclear and time-consuming to analyze manually.	Created monthly trend line charts.	Clearly displays seasonal variations and demand fluctuations.
Decision Support	Manual analysis slows decision-making process.	Added interactive filters (Year, Region, State) and cross-sheet filtering.	Improves decision-making speed through dynamic and user-friendly interaction.